FINANCIAL DATA ANALYSIS

Week 4 Capstone Project Report

Exploratory Data Analysis • Sales Regression • Business Recommendations

Dataset rows	700	**Original columns**	16
Total Sales	118,726,350.26	**Total Profit**	16,893,702.26
Profit Margin	14.23%	**Years**	2013–2014

Prepared using the supplied Financial Data.csv dataset.

1. Executive Summary

This project analyzes 700 records across 16 original columns. Total sales are 118,726,350.26, total profit is 16,893,702.26, total units sold are 1,125,806, and overall profit margin is 14.23%. The highest-sales country is United States of America, the highest-sales product is Paseo, and the highest-sales segment is Government.

2. Objectives

• Clean and explore the dataset using Python and Pandas.

• Perform EDA across countries, products, segments, years and discount bands.

• Train and evaluate a Linear Regression model to predict Sales.

• Design a Power BI dashboard for business KPIs and trends.

• Summarize findings, limitations and recommendations.

3. Dataset and Data Preparation

The source CSV contains 700 rows and 16 columns, including Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month Number, Month Name and Year. The data spans 2013–2014. The original Discount Band field contains 53 missing values. Column names were trimmed, numeric fields converted, dates parsed, exact duplicate rows checked, and missing discount categories marked 'Unknown' rather than assuming they mean no discount. Derived fields include Profit Margin % and Discount Rate %.

4. Key Performance Indicators

Metric	Result
Total Sales	118,726,350.26
Total Profit	16,893,702.26
Units Sold	1,125,806
Total Discounts	9,205,248.24
COGS	101,832,648.00
Profit Margin	14.23%

Country leader: United States of America with sales of 25,029,830.16. Product leader: Paseo with sales of 33,011,143.95. Segment leader: Government with sales of 52,504,260.67. The highest-sales year is 2014. Values are shown without a currency symbol because the source does not specify currency.

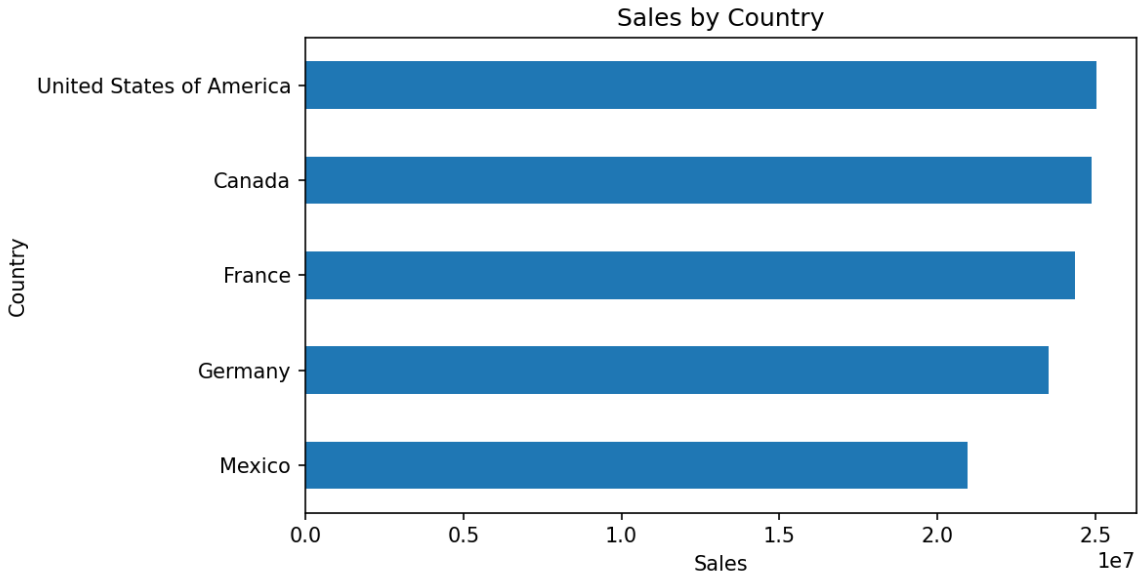


Figure 1. Sales by country.

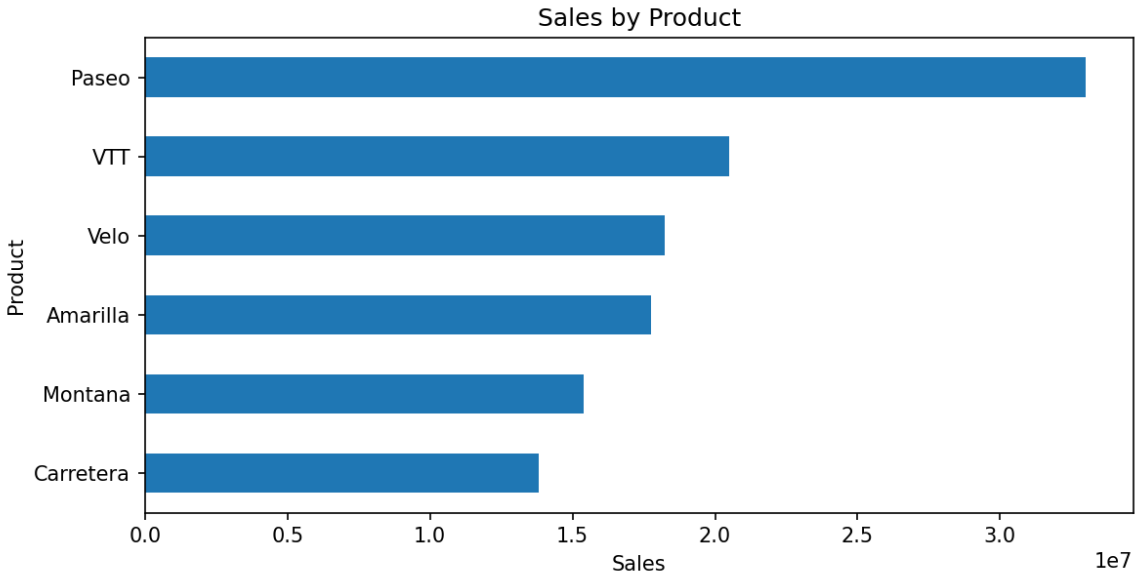


Figure 2. Sales by product.

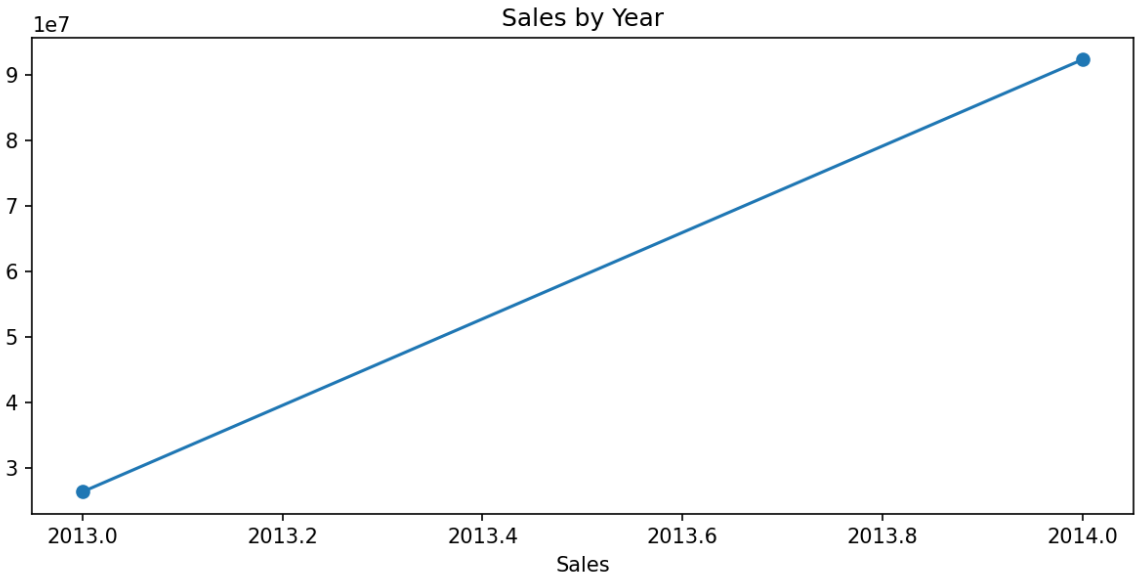


Figure 3. Sales by year.

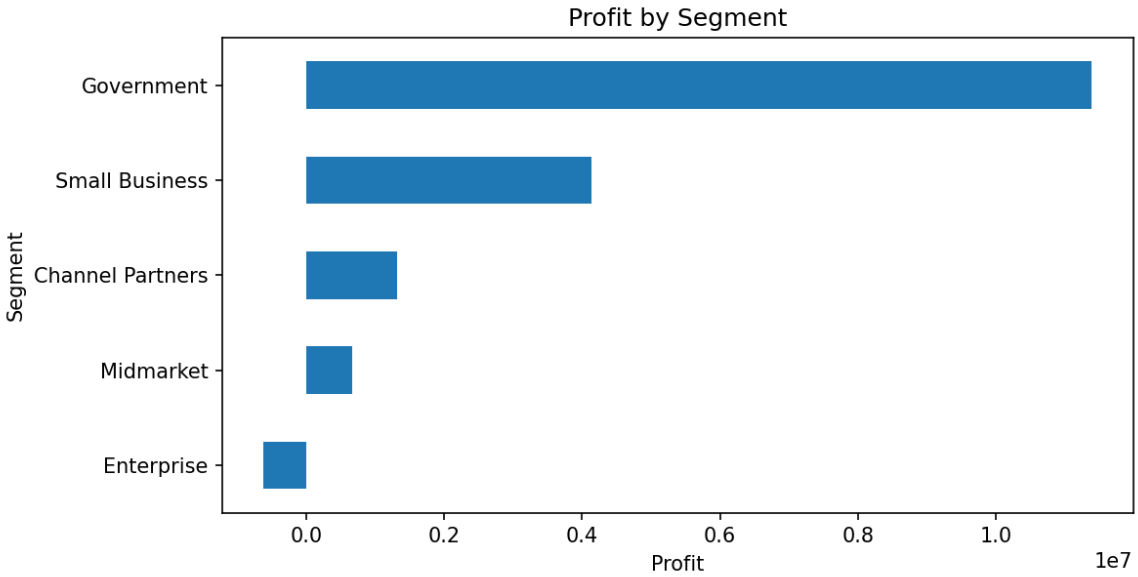


Figure 4. Profit by segment.

5. Sales Regression Model

A baseline Linear Regression model was trained with Sales as the target. Predictors were Units Sold, Manufacturing Price, Sale Price, Discounts, Month Number, Year, Segment, Country, Product and Discount Band. Gross Sales, Profit and COGS were excluded as they are closely related to the sales outcome and could introduce leakage. Numeric missing values were imputed with the median; categorical values were imputed with the most frequent category and one-hot encoded. Data was split into 80% training and 20% testing sets with random_state=42.

Metric	Test result	Meaning
MAE	78,386.63	Mean absolute prediction error
RMSE	109,673.34	Penalizes larger errors
R^2	0.7828	Explained variance on test set

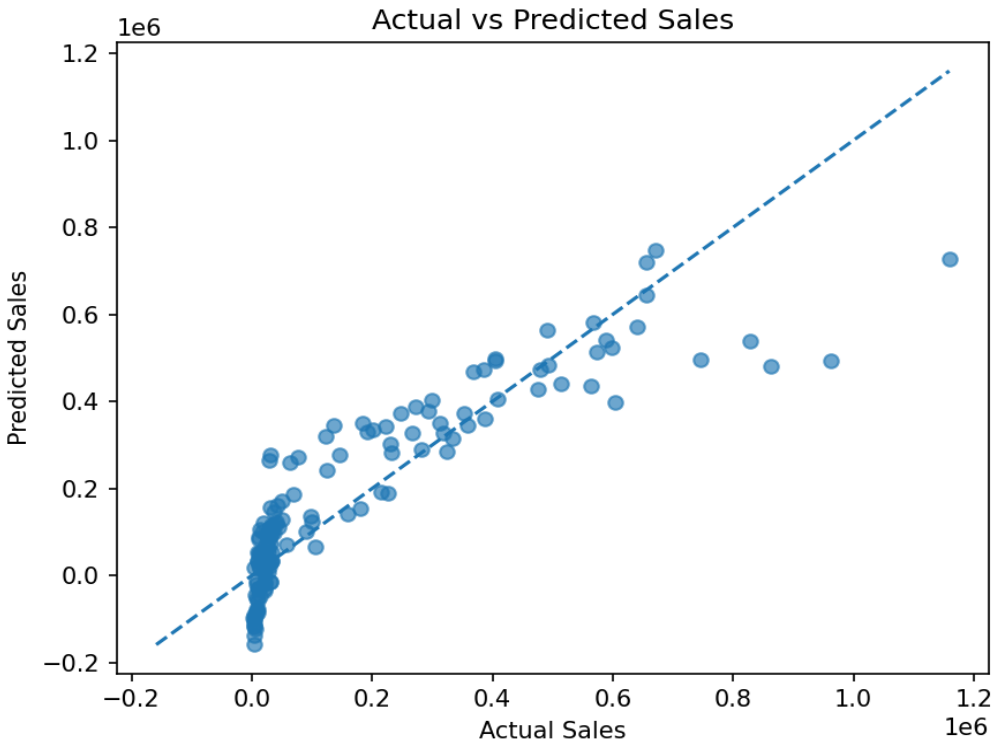


Figure 5. Actual versus predicted sales; points closer to the diagonal represent closer predictions.

These metrics describe a random held-out test set and are not a guarantee of future forecasting accuracy. Because the data is time-based, a chronological holdout and comparison with a simple baseline model are recommended for genuine future-period forecasting.

6. Power BI Dashboard Plan

Recommended title: Financial Performance Dashboard. Include KPI cards for Total Sales, Total Profit, Units Sold, Total Discounts and Profit Margin %. Add charts for Sales by Country, Sales by Product, Sales by Year, Profit by Segment, and Sales/Profit by Discount Band. Add slicers for Year, Country, Segment, Product and Discount Band. Save the native dashboard as Financial_Performance_Dashboard.pbix from Power BI Desktop.

7. Findings and Recommendations

• Investigate the United States of America as the leading sales country, while checking its profit and margin before allocating more resources.

• Compare Paseo's sales with its profit contribution; high sales alone do not prove the strongest profitability.

• Government is the leading segment by sales. Compare segment margins before prioritizing investment.

• Evaluate discounts by considering both sales and profit, not volume alone.

• Investigate year-to-year trends and possible seasonality before setting targets.

• For future sales forecasting, use a chronological test period and compare Linear Regression with baseline and tree-based models.

8. Limitations

The dataset does not specify a currency. Missing Discount Band values are labelled Unknown and should be confirmed with the data owner. This descriptive analysis does not establish causality. A random train/test split may mix dates, so performance may differ when predicting genuinely future periods. The report documents a dashboard plan; the .pbix file must be created and saved separately.

9. Submission Checklist

• financial_data_cleaned.csv — cleaned data.

• Financial_Data_Analysis.ipynb — Python cleaning, EDA and regression code.

• Financial_Data_Analysis.xlsx — Excel analysis workbook.

• Financial_Performance_Dashboard.pbix — Power BI file created in Desktop.

• Financial_Data_Analysis_Report.pdf — this report.

• sales_regression_predictions.csv and sales_regression_metrics.csv — regression outputs.